

POWERFACTORY

PowerFactory 2024

Technical Reference

IEC Wind Turbine Templates

IEC 61400-27-1 WT Type1A 0.5MVA 50Hz,
IEC 61400-27-1 WT Type1A 0.5MVA 60Hz,
IEC 61400-27-1 WT Type1B 0.5MVA 50Hz,
IEC 61400-27-1 WT Type1B 0.5MVA 60Hz,
IEC 61400-27-1 WT Type2 0.5MVA 50Hz,
IEC 61400-27-1 WT Type2 0.5MVA 60Hz,
IEC 61400-27-1 WT Type3A 2.0MW 50Hz,
IEC 61400-27-1 WT Type3A 2.0MW 60Hz,
IEC 61400-27-1 WT Type3B 2.0MW 50Hz,
IEC 61400-27-1 WT Type3B 2.0MW 60Hz,
IEC 61400-27-1 WT Type4A 2.0MW 50Hz,
IEC 61400-27-1 WT Type4A 2.0MW 60Hz,
IEC 61400-27-1 WT Type4B 2.0MW 50Hz,
IEC 61400-27-1 WT Type4B 2.0MW 60Hz

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Publisher:
DIgSILENT GmbH
Heinrich-Hertz-Straße 9
72810 Gomaringen / Germany
Tel.: +49 (0) 7072-9168-0
Fax: +49 (0) 7072-9168-88
info@digsilent.de

Please visit our homepage at:
https://www.digsilent.de

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1 Introduction

This document describes the usage of the IEC 61400-27-1 wind turbine generator (WTG) models in *PowerFactory*. The models are based on [2]. This document will describe the usage of these models in *PowerFactory* and it will highlight details which either deviate or are not specified in [2].

Four different types (seven models) are described in the IEC 61400-27-1 and available in *PowerFactory*:

- Type 1a (WTG with asynchronous generator directly connected to the grid)
- Type 1b (WTG with asynchronous generator directly connected to the grid with UVRT pitch control)
- Type 2 (WTG with asynchronous generator with variable rotor resistance directly connected to the grid)
- Type 3a (WTG with doubly fed induction generator (DFIG))
- Type 3b (WTG with doubly fed induction generator with crowbar)
- Type 4a (WTG connected though full scale power converter)
- Type 4b (WTG connected though full scale power converter with mechanical model)

This documents starts with a general description on the usage for all models in chapter 2. In chapter 4 the specific models and their implementation (if it is different from the standard) are described.

Note: The models which are available in *PowerFactory* as templates are configured with a set of sample parameters, which do not correspond to any IEC recommended settings. Is responsibility of the user to adjust these parameters (i.e. power rating, voltage level, reactive power limits and all parameters of the dynamic models) to represent a specific WTG of a specific manufacturer!

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2 Model Usage

All six models consist of at least two grid elements (either the asynchronous generator *ElmAsm* or the static generator *ElmGenstat*), a terminal and of a composite model which contains all dynamic models and measurement devices. The models are available as templates.

A template can be added to an active project as follows:

- Press the *General Templates* button in the drawing toolbox.
- A list of available templates will be shown. Select from the sub-folder “Wind Turbines” the one you want to use.
- Click into the drawing area. You can see the shape of the selected template attached to the mouse pointer.
- Place the template with another mouse click.
- Close the template window after you finished adding the WTG models.

You may need a configured model several times either in the active project or in another project. It is not possible to use the copy&paste function to replicate the model since this would create only a copy of the grid element (and not of the controller). Also the connection of the measurement devices does not allow to use the copy&paste function. Please use the template function to duplicate the model or to bring it to another project. Follow the steps below to do this:

- Mark all elements belonging to the WTG you want to duplicate. This may also include a transformer. Remind to mark all elements where measurement devices are connected to. The IEC WTG template models are using only the LV bus as measurement location.
- Right click now at the marked elements and select “Define template”.
- Select a meaningful name and press “OK”.
- Confirm in the next step with “Yes” that you want to include external objects (the external object is the composite model with all controllers and measurement devices).
- You can use now the just created template in the active project as described above.

There are some more steps necessary to move the above created template to another project:

- The above create template is stored in the project library in the sub-folder “Templates”. Open this folder in the data manger.
- Right click then at the previously created template and select “Edit”.
- Press in the following window the “Pack” button. This will copy all types of the model to the template.
- The template can now be copied to any other project (to the Template folder of the project library) or exported as PFD file.

3 Model Configuration

This chapter contains information about the configuration of the four different types. At the end the access of the dynamic models is described.

3.1 Type 1 and 2

In the following sub-chapters is the general configuration of type 1 and 2 described like adjusting the rated power or changing the number of parallel machines.

3.1.1 Adjusting Rated Power

The rated power is defined via the type of the asynchronous generator. The type can be accessed via the asynchronous generator element (Figure 3.1). The rated power as well as other attributes such as the inertia of the generator can be changed in the type.

Note: Changing the type will affect all elements using this type. Create a copy of the type to avoid this.

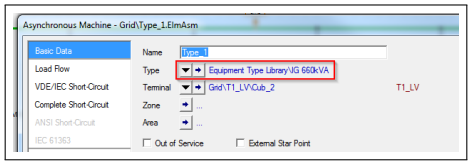


Figure 3.1: Accessing the type of an asynchronous generator

3.1.2 Adjusting Active Power Dispatch

The dispatch of the generator can be entered on the load flow page. The default bus type is “AS” (for asynchronous generator). Using this bus type allows only the input of an active power dispatch. The reactive power will be calculated via the machine parameters entered in the type. The reactive power demand will be compensated with the also available shunt capacitors.

3.1.3 Power-Slip-Characteristic (Type 2 only)

Type 2 needs further configuration of the power dependent slip characteristic. This characteristic can be entered on the load flow page for the generator element. The characteristic “prrdw” has to be entered also in the “Control” model. Read section 3.3 to learn more how to access and edit an array.

Note: The characteristic in the generator is defined via active power in MW (1st column) and slip in % (2nd column).
The characteristic “prrdw” in the “Control” model is defined via slip in % (1st column) and active power in p.u. (2nd column).

3.1.4 Model Aggregation

The WTG can also be used for aggregated representation of several WTGs or even a whole wind farm. The only setting needed for this is the “Number of parallel machines” on the basic data page of the element. The active power output of the generator is then the active power dispatch times the number of parallel machines.

Note: Don't forget to adjust the transformer too.

3.2 Type 3 and 4

In the following sub-chapters is the general configuration of type 3 and 4 described like adjusting the rated power or changing the number of parallel machines.

3.2.1 Adjusting Rated Power

The rated power of the static generator has to be configured on the basic data page of the element itself (i.e. the static generator is not using a type). The rated active power is calculated via the nominal apparent power "sgn" and the power factor "cosn". After changing the rated power the settings on the load flow page should be checked too.

3.2.2 Adjusting Power Dispatch

The dispatch of the generator can be entered on the load flow page. The default bus type is "PQ". Using this bus type allows to enter the active and reactive power dispatch. The reactive power can be limited by the reactive power capability curve which can be also voltage dependent. It is also possible to coordinate the reactive power output of several static generators by using a station controller (*ElmStactrl*).

3.2.3 Model Aggregation

The WTG can also be used for aggregated representation of several WTGs or even a whole wind farm. The only setting needed for this is the "Number of parallel machines" on the basic data page of the element. The active power output of the generator is then the active power dispatch times the number of parallel machines.

Note: Don't forget to adjust the transformer too.

3.3 Dynamic Model Configuration

The dynamic simulation is based in *PowerFactory* on the load flow calculation. This means that the starting point for the dynamic simulation is always a load flow solution. It is therefore important to check that the load flow configuration of the elements is correct. The configuration for the different types is described in chapter 4.1 and 4.2.

The composite model which contains the links to all dynamic model parts (common models) can either be accessed via the generator element in the single line diagram or via the data manager.

- Via Element: Open the element. Go to the basic data page and click at the arrow button after "Plant Model" (type 1 and 2) or at "Model" (type 3 and 4).

- Via Data Manager: Search the corresponding composite model , right click and select “Edit”.

After opening the composite model the different dynamic (common) models can be accessed via a double click in the 2nd column.

The dynamic (common) models can be configured via scalar input parameters. These parameters are displayed after opening a common model on the “General” page.

Some models, such as the “Protection” model contain also arrays (i.e. a list of parameters). The arrays can be found on the second page of the common model. The second page can be accessed by selecting the tab “Advanced 1”.

To change the dimension of an array simply double click at the cell in the second column. This will open an *IntMat* object as shown in Figure 3.2. Right clicking allows then to add more rows to the array.

Note: The the first column of an array has to be in an ascending order!
It is not allowed to change the name of the *IntMat* object.

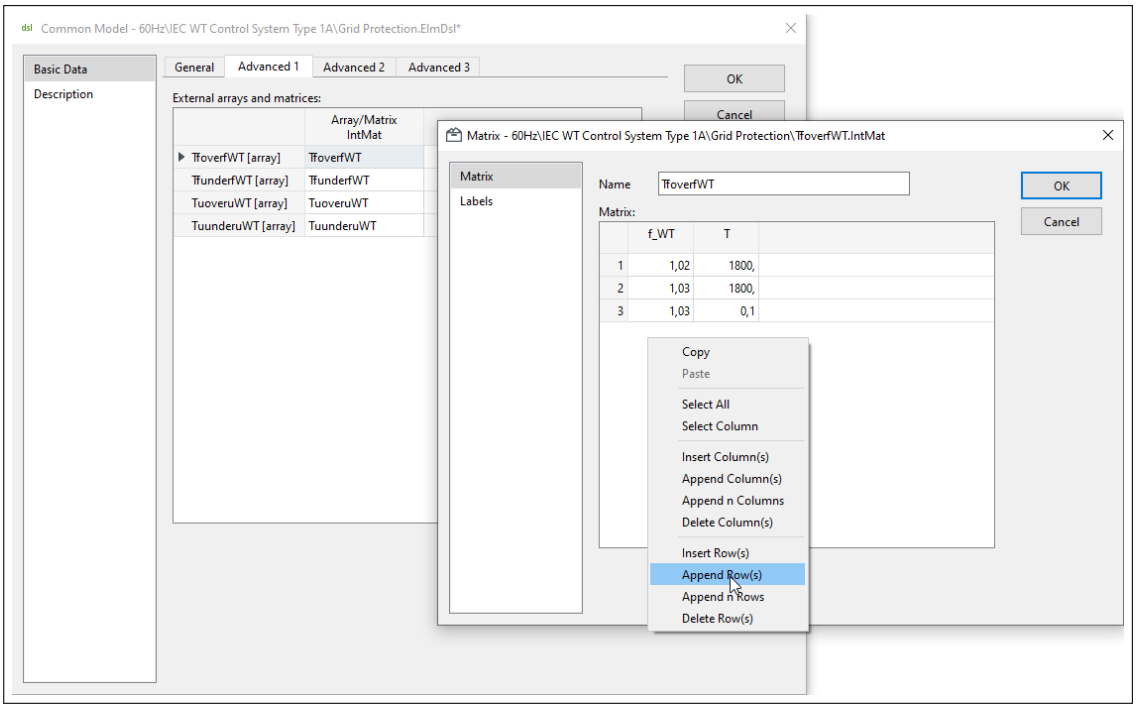


Figure 3.2: Array input (IntMat object)

4 Model Documentation

The following sub-sections will describe the seven different types implemented according to IEC 61400-27-1. All parts of the dynamic models are listed in table 4.1. The seven different models are split in two groups and are described in 4.1 and 4.2. Please read chapter 3.3 to learn more about the configuration of the dynamic model in *PowerFactory*.

Blocks	1A	1B	2	3A	3B	4A	4B
Aerodynamic (Const. Aero Torque)	X						
Aerodynamic (Two-dimensional aero model)				X[a]	X[a]		
Aerodynamic (One-dimensional aero model)				X[a]	X[a]		
Mechanical (Two mass) modified for ElmAsm	X	X	X				
Mechanical (Two mass)				X	X		X
Generator System (Induction Gen.)	X	X	X				
Generator System (Type 3A Gen. Set Model)				X			
Generator System (Type 3B Gen. Set Model)					X		
Generator System (Type 4 Gen. Set Model)						X	X
Reference frame rotation model (PLL)				X	X	X	X
Control (Pitch control power model)		X	X				
Control (Rotor resistance control model)			X				
Control (P control type 3)				X	X		
Control (P control type 4A)						X	
Control (P control type 4B)							X
Control (Q control mode)				X	X	X	X
Control (Current limitation model)				X	X	X	X
Constant Q limitation model				X[b]	X[b]	X[b]	X[b]
QP and QU limitation model				X[b]	X[b]	X[b]	X[b]
Control (Pitch angle control model)				X	X		
Grid Protection	X	X	X	X	X	X	X
Grid Measurement	X	X	X	X	X	X	X

Table 4.1: Block Overview

[a] Only one of the two aero dynamic models can be used at the same time
[b] Only one of the two limitation models can be used at the same time

4.1 Type 1A, 1B and 2

The type 1A, 1B and 2 WTG is in the network represented via the asynchronous generator element and a shunt capacitor which is used to compensate the reactive power demand of the generator (Figure 4.1). The initial settings for the load flow calculation and dynamic simulation are done in this element. The configuration of the grid element is described in section 3.1.

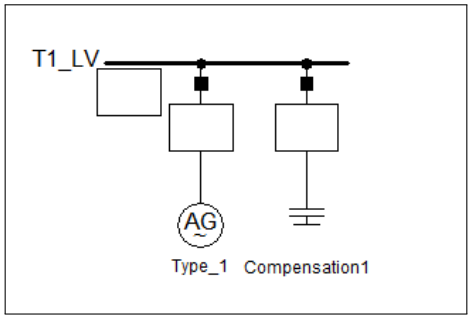


Figure 4.1: Type 1 network representation

4.2 Type 3A, 3B, 4A and 4B

The type 3A/B 4A/B WTG is in the network represented via the static generator element (Figure 4.2). The initial settings for the load flow calculation and dynamic simulation are done in this element. The configuration of the grid element is described in section 3.2.

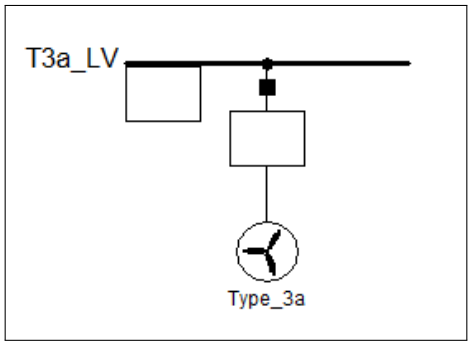


Figure 4.2: Type 3a network representation

4.2.1 Reference Frame Rotation Model (PLL)

The reference frame rotation model is implemented as specified in the standard using the measured real and imaginary components of the voltage. These magnitudes are in *PowerFactory* obtained from the voltage measurement device *StaVmea* which is considered to monitor at generator terminals, this is at the transformer LV side. Should be desired to monitor the voltage at a the HV side then the user has to consider the following modifications in the model:

- A second voltage measurement device is required, such that one monitors at the LV side (required for the PLL) and the second one can be located at a different point. This requires also a modification in the frame.

The PLL is used to calculate the voltage angle via an algebraic approach. This can lead in weak network situations to stability problems. In case of such stability problems there are two possible solutions:

- Increasing the parameter “uPLL1” to a higher value. This will enable a first order lag for the voltage earlier or permanent (in case of a value significant higher than 1).
- The second possibility is to use the build in PLL element (*ElmPhi_pll*) of *PowerFactory*. Read the technical reference paper [1] for further details about the PLL of *PowerFactory*.

4.3 Frequency Measurement

All models use the frequency as input to the grid measurement model, which send the filtered measured signal to the protection model and in Type 2 also to the Rotor Resistance Control model. The frequency signal is taken from the output of the voltage measurement device *StaVmea* [3].

5 References

[1] *Technical Reference "Phase Locked Loop"*. DIgSILENT GmbH, Gomaringen, Germany, 2016.

[2] IEC 61400-27-1 Electrical Simulation Models for Wind Power Generation; Edition 2.0, 2020.

[3] *Technical Reference "Voltage Measurement Device"*. DIgSILENT GmbH, Gomaringen, Germany, 2023.

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